

Task: Fine-Tuning vs. Training a Small Language Model from Scratch

AI tools: Allowed. However, you must be able to explain and defend every line of code and every decision in a live technical discussion. That discussion carries most of the weight in our evaluation.

The task

Build and compare two language-modeling approaches using the provided document:

Track 1 — Fine-tune a pretrained model. Choose any pretrained language model and fine-tune it on content derived from the document. The model, the fine-tuning method, and how you prepare the data are entirely your decisions — be ready to justify them.

Track 2 — Train a small language model from scratch. Design and implement your own small model (your own architecture and training code) and train it on the document text. It only needs to be a base language model that can plausibly complete sentences and passages in the document's domain — no instruction-following or question-answering ability is required.

Then compare the two approaches and state what you would recommend in a real production setting, and why.

How you prepare the data, split it, evaluate the models, and manage compute is deliberately left open. Those decisions — and your reasoning for them — are a large part of what we are assessing.

Expected outcome

- Two trained models (fine-tuned and from-scratch) with working, reproducible pipelines
- An evaluation you designed yourself that compares them meaningfully
- A clear analysis of what each approach can and cannot do

Deliverables

1. Code / notebook(s) that reproduce your results end-to-end.
2. A short write-up (1–2 pages).

Your deliverables **must include**, at minimum:

- **Your key choices, each with reasoning** — including base model selection, the **tokenizer choice** for your from-scratch model (train your own vs. reuse a pretrained one, and vocabulary size), data preparation, and the main hyperparameters.
- **Training and validation loss curves for both tracks**, with your interpretation of what they show.
- **Fine-tuning specifics:** the method you used (full fine-tuning, LoRA/QLoRA, or another approach), which parameters were actually trained, and why that method fits your compute.

- A comparison of the two approaches and your production recommendation.

The deliverables must be emailed to the email id from which you received this task.

Practical notes

- Google Colab (free tier) is sufficient for this task and is our suggested environment. You may use any environment you prefer, as long as the results are reproducible and you can walk us through them.
- Keep model sizes and training runs modest. Scoping the work to your compute is part of the task.